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Discrete Structures & Theory of Logic

BCS303

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** CSE, IT, CSE-AIML, CSE-DS, AIDS | **Semester:** 3 | **Credits:** 3

L-T-P: 3-0-0 | **Scheme:** AKTU B.Tech 2nd year, 2023-24 onwards

UNIT 1 Sets, Relations and Lattices

UNIT 2 Functions and Boolean Algebra

UNIT 3 Theory of Logic

UNIT 4 Algebraic Structures

UNIT 5 Graphs and Combinatorics

Original revision notes written for the AKTU syllabus. Every unit has the same parts:

1. **2-mark definitions**: short answers you can write word for word in Section A.
2. **Core concepts**: the results, formulas and tables you need.
3. **Solved examples**: the proofs and problems that keep appearing (Hasse diagrams, K-maps, inference, cosets, planarity).
4. **Results to remember**: the theorems and formulas to write from memory.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (3 hours):** Units 1 and 2. Draw Hasse diagrams for D12, D30 and similar sets, and solve two K-maps (3 and 4 variables).
- **Afternoon (3 hours):** Units 3 and 4. Practise one inference proof in propositional logic and one in predicate logic; learn group definitions and Lagrange's theorem with proof.
- **Evening (2 hours):** Unit 5 (Euler/Hamiltonian, planarity, colouring, pigeonhole), then the "Last-hour checklist".
- **In the exam:** state the definition before every proof. Write the rule name (Modus Ponens, Hypothetical Syllogism...) next to each inference step. Draw every graph and Hasse diagram neatly.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit.

UNIT 1 Sets, Relations and Lattices

✍️ 2-mark definitions

- **Set:** a well-defined collection of distinct objects. **Power set P(A):** the set of all subsets of A; if $|A| = n$ then $|P(A)| = 2^n$.
- **Cartesian product:** $A \times B = \{(a, b) : a \in A, b \in B\}$; $|A \times B| = |A| \cdot |B|$.
- **Relation:** a subset R of $A \times B$. A relation on A is a subset of $A \times A$.
- **Reflexive:** $(a, a) \in R$ for every $a \in A$. **Irreflexive:** $(a, a) \notin R$ for every a.
- **Symmetric:** $(a, b) \in R \Rightarrow (b, a) \in R$. **Antisymmetric:** $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$. **Asymmetric:** $(a, b) \in R \Rightarrow (b, a) \notin R$.
- **Transitive:** $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$.
- **Equivalence relation:** reflexive, symmetric and transitive. It splits the set into disjoint **equivalence classes** (a partition).
- **Partial order (POSET):** a relation that is reflexive, antisymmetric and transitive, written $(A, \leq)$.
- **Total (linear) order / chain:** a partial order in which every two elements are comparable.
- **Hasse diagram:** a drawing of a POSET with loops and transitive edges removed, smaller elements placed lower.
- **Composite relation:** $R \circ S = \{(a, c) : (a, b) \in R$ and $(b, c) \in S$ for some $b\}$.
- **Lattice:** a POSET in which every pair of elements has a least upper bound (join, $a \vee b$) and a greatest lower bound (meet, $a \wedge b$).

💡 Core concepts

Set identities: De Morgan $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$; distributive $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$; inclusion–exclusion $|A \cup B| = |A| + |B| - |A \cap B|$ and $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$.

Counting relations on a set with n elements

Kind	Number
All relations	$2^{(n^2)}$
Reflexive	$2^{(n^2 - n)}$
Symmetric	$2^{(n(n+1)/2)}$
Reflexive and symmetric	$2^{(n(n-1)/2)}$
Antisymmetric	$2^n \cdot 3^{(n(n-1)/2)}$

Matrix test for properties (M = relation matrix): reflexive if every diagonal entry is 1; symmetric if $M = M^T$; antisymmetric if no $i \neq j$ has $M_{ij} = M_{ji} = 1$; transitive if $M \odot M$ (Boolean product) $\leq M$.

Closures: reflexive closure $R \cup \Delta$; symmetric closure $R \cup R^{-1}$; transitive closure $R^+ = R \cup R^2 \cup \ldots \cup R^n$ (computed with Warshall's algorithm). **Recursive definition of a relation:** give a base case and a rule that

builds new pairs from old ones (e.g. transitive closure).

POSET terms: maximal / minimal element (nothing above / below it); greatest / least element (above / below everything); upper bound, lower bound; **lub (supremum)**, **glb (infimum)**.

Types of lattices

Lattice	Condition
Bounded	Has a least element 0 and a greatest element 1
Complemented	Bounded, and every a has a′ with a ∨ a′ = 1, a ∧ a′ = 0
Distributive	a ∧ (b ∨ c) = (a ∧ b) ∨ (a ∧ c) for all a, b, c
Modular	a ≤ c ⇒ a ∨ (b ∧ c) = (a ∨ b) ∧ c
Complete	Every subset has a lub and a glb (every finite lattice is complete)

- In a **distributive** lattice, complements are unique when they exist.
- The **diamond M_3** (five elements, three in the middle) is modular but not distributive. The **pentagon N_5** is neither modular nor distributive. A lattice is distributive exactly when it contains neither as a sublattice.
- A complemented distributive lattice is a **Boolean algebra**.

🧮 Solved examples

Example 1 (equivalence relation). On A = {1, 2, 3}, R = {(1,1), (2,2), (3,3), (1,2), (2,1)}.

- Reflexive: all (a, a) present ✔. Symmetric: (1,2) and (2,1) ✔. Transitive: (1,2),(2,1) ⇒ (1,1) ✔ and (2,1), (1,2) ⇒ (2,2) ✔.
- So R is an equivalence relation, with classes **{1, 2}** and **{3}**.

Example 2 (Hasse diagram of D_{12} under divisibility). D_{12} = {1, 2, 3, 4, 6, 12}.



Edges: 1–2, 1–3, 2–4, 2–6, 3–6, 4–12, 6–12. Join = lcm, meet = gcd, so it is a lattice with 0 = 1 and 1 = 12.

- **Distributive:** yes (gcd and lcm distribute).
- **Complemented: no.** A complement of 2 would need gcd(2, x) = 1 and lcm(2, x) = 12; no x in D_{12} works. (D_{30} = {1, 2, 3, 5, 6, 10, 15, 30} *is* complemented, e.g. 2′ = 15, so D_{30} is a Boolean algebra.)

Example 3 (composite relation). R = {(1,2), (2,3)} and S = {(2,4), (3,1)}. R∘S: (1,2) with (2,4) gives (1,4); (2,3) with (3,1) gives (2,1). **R∘S = {(1,4), (2,1)}**.

Results to remember

```
|P(A)| = 2^n          |A × B| = |A|·|B|
Equivalence = reflexive + symmetric + transitive        → partition into classes
Partial order = reflexive + antisymmetric + transitive  → Hasse diagram
Lattice: every pair has lub (join ∨) and glb (meet ∧)
Divisibility lattice Dn: join = lcm, meet = gcd
Dn is a Boolean algebra exactly when n is square-free (e.g. D30, not D12)
```

⭐ Important questions

1. Define equivalence relation and show a given relation is one; find its equivalence classes. *Check all three properties explicitly.*
2. Draw the Hasse diagram of a POSET (e.g. divisors of n) and find maximal, minimal, greatest, least elements, upper and lower bounds.
3. Define lattice and its types (bounded, complemented, distributive, modular). Check whether a given lattice is distributive/complemented. *Use M_3/N_5 as counter-examples.*
4. Prove that in a distributive lattice the complement of an element, if it exists, is unique.
5. Find the transitive closure of a relation using Warshall's algorithm.

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UNIT 2 Functions and Boolean Algebra

✍️ 2-mark definitions

- **Function f: A → B:** a relation that gives every element of A exactly one image in B. A is the domain, B the codomain, f(A) the range.
- **One-one (injective):** $f(a_1) = f(a_2) \Rightarrow a_1 = a_2$. **Onto (surjective):** every $b \in B$ has a pre-image. **Bijective:** both one-one and onto; only bijections have inverses.
- **Composition:** $(g \circ f)(x) = g(f(x))$. It is associative but not commutative in general.
- **Inverse function f^{-1}:** $f^{-1}(b) = a$ exactly when $f(a) = b$.
- **Identity, constant, floor/ceiling functions:** $I(x) = x$; $f(x) = c$; $\lfloor x \rfloor$ greatest integer $\leq x$, $\lceil x \rceil$ smallest integer $\geq x$.
- **Growth of functions:** comparing how fast functions grow using O, Ω and Θ notation.
- **Boolean algebra:** a set B with operations +, · and ′, and elements 0 and 1, satisfying the Huntington axioms (closure, identity, commutative, distributive, complement).
- **Minterm / maxterm:** a product (sum) term that contains every variable exactly once, complemented or not.
- **SOP / POS:** sum of products / product of sums forms of a Boolean function.
- **Karnaugh map:** a grid of minterms arranged in Gray-code order so adjacent cells differ in one variable, used to simplify Boolean functions.

💡 Core concepts

Counting functions from A ($|A| = m$) to B ($|B| = n$)

Kind	Number
All functions	n^m
One-one ($m \leq n$)	$n! / (n - m)!$
Onto ($m \geq n$)	$\Sigma_{k=0}^{n} (-1)^k C(n, k) (n - k)^m$
Bijections ($m = n$)	$n!$

Growth of functions: $f(n) = O(g(n))$ if $f(n) \leq c \cdot g(n)$ for $n \geq n_0$; Ω for ≥; Θ when both hold. Order: $1 < \log n < n < n \log n < n^2 < 2^n < n!$.

Boolean algebra laws

Law	Form
Identity	$A + 0 = A$, $A \cdot 1 = A$
Null (domination)	$A + 1 = 1$, $A \cdot 0 = 0$
Idempotent	$A + A = A$, $A \cdot A = A$

Law	Form
Complement	A + A′ = 1, A · A′ = 0
Involution	(A′)′ = A
Absorption	A + AB = A, A(A + B) = A
Useful	A + A′B = A + B, (A + B)(A + C) = A + BC
De Morgan	(A + B)′ = A′B′, (AB)′ = A′ + B′
Consensus	AB + A′C + BC = AB + A′C

Duality principle: swapping + with · and 0 with 1 in any valid identity gives another valid identity.

K-map rules: group 1s in rectangles of size 1, 2, 4, 8 (powers of 2); make groups as large as possible; groups may overlap and wrap around edges; every 1 must be covered; use the fewest groups. **Don't-care** cells (X) may be used if they help make bigger groups.

🧮 Solved examples

Example 1 (composition and inverse). f(x) = 2x + 1 and g(x) = x^2 on ℝ.

- (g∘f)(x) = g(2x + 1) = **$(2x + 1)^2$**. (f∘g)(x) = f(x^2) = **$2x^2 + 1$**. They differ, so composition is not commutative.
- f is a bijection: y = 2x + 1 ⇒ **$f^{-1}(y) = (y - 1)/2$**.

Example 2 (one-one and onto proof). f: ℝ → ℝ, f(x) = 3x − 5.

- One-one: 3a − 5 = 3b − 5 ⇒ a = b ✔.
- Onto: for any y, x = (y + 5)/3 gives f(x) = y ✔. So f is bijective.

Example 3 (3-variable K-map). F(A, B, C) = Σm(0, 2, 4, 5, 6).

```
        B'C'  B'C   BC   BC'
A'       1     0     0    1        (m0, m1, m3, m2)
A        1     1     0    1        (m4, m5, m7, m6)
```

- Group {0, 2, 4, 6} (all cells with C = 0) → **C′**.
- Group {4, 5} → **AB′**.
- **F = C′ + AB′**.

Example 4 (4-variable K-map). F(A, B, C, D) = Σm(0, 1, 2, 5, 8, 9, 10).

- Quad {0, 1, 8, 9} → **B′C′**. Quad {0, 2, 8, 10} (the four corners) → **B′D′**. Pair {1, 5} → **A′C′D**.
- **F = B′C′ + B′D′ + A′C′D**.

Example 5 (algebraic simplification). Simplify AB + A′C + BC.

- BC = BC(A + A′) = ABC + A′BC. Then AB + ABC = AB and A′C + A′BC = A′C.
- **Result: AB + A′C** (consensus theorem).

Results to remember

```
Functions A→B, |A| = m, |B| = n:   all = n^m;   one-one = n!/(n-m)!;   bijections = n!
Only bijections have inverses;   (g∘f)⁻¹ = f⁻¹ ∘ g⁻¹
A + A'B = A + B          (A + B)(A + C) = A + BC
De Morgan: (A + B)' = A'B',  (AB)' = A' + B'
K-map groups: 1, 2, 4, 8 cells; bigger is better; wrap around edges
```

⭐ Important questions

1. Define one-one, onto and bijective functions; check a given function and find its inverse. *Prove both properties step by step.*
2. Find the number of one-one/onto functions between two finite sets.
3. State and prove De Morgan's laws and other theorems of Boolean algebra (absorption, consensus).
4. Simplify a Boolean function using a K-map (3 or 4 variables, maybe with don't-cares). *Draw the map, circle the groups, write the term for each.*
5. Express a function in canonical SOP and POS forms.

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UNIT 3 Theory of Logic

✍️ 2-mark definitions

- **Proposition:** a declarative sentence that is either true or false, but not both.
- **Connectives:** negation ¬, conjunction ∧, disjunction ∨, conditional →, biconditional ↔.
- **Tautology:** a compound proposition that is true for every assignment of truth values. **Contradiction:** false for every assignment. **Contingency:** neither.
- **Satisfiable:** true for at least one assignment.
- **Logical equivalence (p ≡ q):** p ↔ q is a tautology.
- **Converse, inverse, contrapositive of p → q:** q → p; ¬p → ¬q; ¬q → ¬p. Only the contrapositive is equivalent to p → q.
- **Argument / valid argument:** premises leading to a conclusion; valid if the conclusion is true whenever all premises are true.
- **Predicate:** a statement containing variables, which becomes a proposition when values are given, e.g. P(x): "x > 3".
- **Quantifiers:** universal ∀x ("for all x") and existential ∃x ("there exists x").
- **Well-formed formula (WFF):** an expression built correctly from predicates, connectives and quantifiers according to the formation rules.
- **Free vs bound variable:** a variable is bound if it lies in the scope of a quantifier on it; otherwise it is free.

💡 Core concepts

Truth table of connectives

p	q	¬p	p ∧ q	p ∨ q	p → q	p ↔ q
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Key equivalences (algebra of propositions)

- p → q ≡ ¬p ∨ q ≡ ¬q → ¬p
- p ↔ q ≡ (p → q) ∧ (q → p)
- De Morgan: ¬(p ∧ q) ≡ ¬p ∨ ¬q; ¬(p ∨ q) ≡ ¬p ∧ ¬q
- Distributive, absorption, idempotent, double negation, as in Boolean algebra.
- **Normal forms:** DNF (sum of products) and CNF (product of sums); PDNF/PCNF use full minterms/maxterms.

Rules of inference

Rule	Form
Modus Ponens	$p, p \to q \vdash q$
Modus Tollens	$\neg q, p \to q \vdash \neg p$
Hypothetical Syllogism	$p \to q, q \to r \vdash p \to r$
Disjunctive Syllogism	$p \vee q, \neg p \vdash q$
Addition	$p \vdash p \vee q$
Simplification	$p \wedge q \vdash p$
Conjunction	$p, q \vdash p \wedge q$
Resolution	$p \vee q, \neg p \vee r \vdash q \vee r$
Constructive dilemma	$(p \to q) \wedge (r \to s), p \vee r \vdash q \vee s$

Rules **P** (introduce a premise) and **T** (use a tautology/earlier lines) and **CP** (conditional proof: to prove $p \to q$, assume p and derive q) are used in formal proofs.

Predicate logic

- Negation: $\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$; $\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$.
- $\forall$ distributes over $\wedge$: $\forall x\, (P(x) \wedge Q(x)) \equiv \forall x\, P(x) \wedge \forall x\, Q(x)$. $\exists$ distributes over $\vee$.
- **Inference rules:** Universal Instantiation (US): $\forall x\, P(x) \vdash P(c)$; Universal Generalization (UG): $P(c)$ for arbitrary $c \vdash \forall x\, P(x)$; Existential Instantiation (ES): $\exists x\, P(x) \vdash P(c)$ for some new c; Existential Generalization (EG): $P(c) \vdash \exists x\, P(x)$.
- Translating English: "All students are clever" → $\forall x\, (S(x) \to C(x))$. "Some students are clever" → $\exists x\, (S(x) \wedge C(x))$. (Use $\to$ with $\forall$ and $\wedge$ with $\exists$.)

🧮 Solved examples

Example 1 (tautology by truth table). Show $[(p \to q) \wedge p] \to q$ is a tautology.

p	q	$p \to q$	$(p \to q) \wedge p$	$[(p \to q) \wedge p] \to q$
T	T	T	T	**T**
T	F	F	F	**T**
F	T	T	F	**T**
F	F	T	F	**T**

The last column is all T, so it is a **tautology** (this is Modus Ponens).

Example 2 (equivalence without truth table). Show $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$.

- $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$ (De Morgan)
- $\equiv \neg p \wedge (p \vee \neg q)$ (De Morgan, double negation)
- $\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$ (distributive)
- $\equiv F \vee (\neg p \wedge \neg q) \equiv \mathbf{\neg p \wedge \neg q}$ ✔

Example 3 (validity using inference rules). Premises: p → q, q → r, ¬r. Prove ¬p.

Step	Statement	Reason
1	p → q	Premise
2	q → r	Premise
3	p → r	Hypothetical Syllogism (1, 2)
4	¬r	Premise
5	**¬p**	Modus Tollens (3, 4)

Example 4 (English argument). "If it rains, the match is cancelled. If the match is cancelled, we go home. It is raining. So we go home." Let r = rains, c = match cancelled, h = go home. Premises r → c, c → h, r. Then r → h (Hypothetical Syllogism) and with r, **h** follows by Modus Ponens. The argument is valid.

Example 5 (predicate inference). "All men are mortal. Socrates is a man. Therefore Socrates is mortal."

Step	Statement	Reason
1	∀x (M(x) → D(x))	Premise
2	M(s) → D(s)	Universal Instantiation (1)
3	M(s)	Premise
4	**D(s)**	Modus Ponens (2, 3)

Results to remember

```
p → q ≡ ¬p ∨ q ≡ ¬q → ¬p (contrapositive)
Only p → q is false: T → F
¬∀x P(x) ≡ ∃x ¬P(x)        ¬∃x P(x) ≡ ∀x ¬P(x)
∀ goes with →,  ∃ goes with ∧
MP: p, p→q ⊢ q      MT: ¬q, p→q ⊢ ¬p      HS: p→q, q→r ⊢ p→r      DS: p∨q, ¬p ⊢ q
```

⭐ Important questions

1. Construct a truth table and decide whether a compound proposition is a tautology, contradiction or contingency.
2. Prove a logical equivalence using laws (without a truth table). *Name the law at each step.*
3. Test the validity of an argument using rules of inference (including English arguments). *Table with step, statement, reason.*
4. Explain quantifiers with examples; translate English sentences into predicate logic and negate them.
5. Prove a conclusion using predicate inference rules (US, UG, ES, EG).

6. Obtain the PDNF and PCNF of a given formula.

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UNIT 4 Algebraic Structures

✍️ 2-mark definitions

- **Binary operation:** a rule * that combines two elements of a set S into an element of S (closure).
- **Algebraic structure:** a set with one or more binary operations, written (S, *).
- **Semigroup:** closed and associative. **Monoid:** semigroup with an identity element.
- **Group:** a set G with * that is **closed, associative, has an identity, and every element has an inverse**. **Abelian group:** a group that is also commutative.
- **Order of a group $|G|$:** number of elements. **Order of an element a:** the smallest positive n with $a^n = e$.
- **Subgroup:** a subset H of G that is itself a group under the same operation. Test: H non-empty and $a, b \in H \Rightarrow a \cdot b^{-1} \in H$.
- **Cyclic group:** a group generated by one element a: $G = \langle a \rangle = \{a^n\}$. Every cyclic group is abelian.
- **Coset:** for a subgroup H and $a \in G$, the left coset $aH = \{ah : h \in H\}$ and right coset Ha.
- **Normal subgroup:** H with $aH = Ha$ for every $a \in G$ (equivalently $aHa^{-1} = H$).
- **Permutation group:** a group of bijections of a set onto itself under composition. **Symmetric group S_n:** all n! permutations of n objects.
- **Homomorphism:** a map $f: G \to G'$ with $f(ab) = f(a)f(b)$. **Isomorphism:** a bijective homomorphism. **Kernel:** $\{a : f(a) = e'\}$, always a normal subgroup.
- **Ring:** $(R, +, \cdot)$ where $(R, +)$ is an abelian group, $\cdot$ is associative, and $\cdot$ distributes over $+$.
- **Field:** a commutative ring with unity where every non-zero element has a multiplicative inverse.

💡 Core concepts

Hierarchy: groupoid (closure) → semigroup (+ associativity) → monoid (+ identity) → group (+ inverses) → abelian group (+ commutativity).

Examples to quote

Structure	Example
Semigroup, not monoid	$\mathbb{N}$ without 0, +
Monoid, not group	$(\mathbb{N}, +)$ with 0 or $(\mathbb{Z}, \times)$
Abelian group	$(\mathbb{Z}, +)$, $(\mathbb{Q}\setminus\{0\}, \times)$, $(\mathbb{Z}_n, +_n)$
Non-abelian group	S_3 (6 permutations), invertible n×n matrices
Cyclic group	$\{1, -1, i, -i\}$ under ×, generated by i
Ring, not field	$(\mathbb{Z}, +, \times)$
Field	$\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and $\mathbb{Z}_p$ for prime p

Basic group properties: identity is unique; each element has a unique inverse; $(ab)^{-1} = b^{-1}a^{-1}$; cancellation laws hold.

Lagrange's theorem: if H is a subgroup of a finite group G, then **$|H|$ divides $|G|$**.

- Proof idea: the left cosets of H partition G; each coset has exactly $|H|$ elements; so $|G| = k \cdot |H|$ where k is the number of cosets (the index $[G : H]$).
- Consequences: the order of every element divides $|G|$; a group of prime order is cyclic; a^$|G|$ = e.

Cyclic group facts: a cyclic group of order n has exactly one subgroup for each divisor of n, and $\varphi(n)$ generators. In $\mathbb{Z}_n$, k is a generator exactly when $\gcd(k, n) = 1$.

Permutations: written in two-line or cycle notation. A permutation is **even** or **odd** by the number of transpositions in it; the even permutations form the alternating group A_n (order $n!/2$), a normal subgroup of S_n.

Ring properties: a ring may have zero divisors ($a \cdot b = 0$ with $a, b \neq 0$, e.g. $2 \cdot 3 = 0$ in $\mathbb{Z}_6$). An **integral domain** is a commutative ring with unity and no zero divisors. Every field is an integral domain; every finite integral domain is a field.

🧮 Solved examples

Example 1 (show a group). Show $G = \{1, -1, i, -i\}$ is a group under multiplication.

- Closure: every product is in G (check the table below). Associativity: inherited from complex multiplication. Identity: 1. Inverses: $1^{-1} = 1$, $(-1)^{-1} = -1$, $i^{-1} = -i$, $(-i)^{-1} = i$.
- It is **abelian** and **cyclic**: $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, so $G = \langle i \rangle$ (also $\langle -i \rangle$).

×	1	−1	i	−i
1	1	−1	i	−i
−1	−1	1	−i	i
i	i	−i	−1	1
−i	−i	i	1	−1

Example 2 (subgroups and cosets in $\mathbb{Z}_6$). $G = (\mathbb{Z}_6, +_6)$, $|G| = 6$.

- Subgroups (orders divide 6): {0}, **{0, 3}**, **{0, 2, 4}**, $\mathbb{Z}_6$.
- Cosets of $H = \{0, 3\}$: $0 + H = \{0, 3\}$, $1 + H = \{1, 4\}$, $2 + H = \{2, 5\}$. Three cosets, each of size 2: $3 \times 2 = 6$ ✔ (Lagrange). Index $[G : H] = 3$.
- Generators of $\mathbb{Z}_6$: 1 and 5 (gcd with 6 is 1).

Example 3 (permutation product). $\sigma = (1\ 2\ 3)$ and $\tau = (1\ 2)$ in S_3. Compute $\sigma\tau$, applying τ first.

- $1 \to \tau \to 2 \to \sigma \to 3$, so $1 \to 3$. $2 \to 1 \to 2$, so $2 \to 2$. $3 \to 3 \to 1$, so $3 \to 1$.
- **$\sigma\tau = (1\ 3)$**. Doing $\tau\sigma$ instead gives $(2\ 3)$, so S_3 is **not abelian**.

Example 4 (homomorphism). $f: (\mathbb{Z}, +) \to (\mathbb{Z}_n, +_n)$, $f(x) = x \bmod n$. $f(a + b) = (a + b) \bmod n = f(a) +_n f(b)$, so f is a homomorphism. Kernel = {multiples of n} = $n\mathbb{Z}$.

Results to remember

```
Group = closure + associative + identity + inverse   (abelian: + commutative)
Subgroup test: H ≠ ∅ and a, b ∈ H ⇒ ab⁻¹ ∈ H
Lagrange: |H| divides |G|;  order of any element divides |G|;  prime order ⇒ cyclic
Generators of Zn: k with gcd(k, n) = 1;  number = φ(n)
|Sn| = n!,  |An| = n!/2
Zp is a field ⇔ p is prime
```

⭐ Important questions

1. Define group, abelian group, cyclic group; show a given set and operation forms a group. *Check all four axioms; draw the composition table.*
2. State and prove Lagrange's theorem. Give its corollaries.
3. Find all subgroups and the cosets of a subgroup of a given group (e.g. $\mathbb{Z}_n$). *List cosets and verify Lagrange.*
4. Define normal subgroup and homomorphism; prove the kernel of a homomorphism is a normal subgroup.
5. Prove that every cyclic group is abelian; find generators of a cyclic group.
6. Define ring, integral domain and field, with examples; show $\mathbb{Z}_p$ is a field when p is prime.

NEXT ENG _sheerat_

UNIT 5 Graphs and Combinatorics

✍️ 2-mark definitions

- **Graph G = (V, E):** a set of vertices and a set of edges joining pairs of vertices.
- **Simple graph:** no loops and no parallel edges. **Multigraph:** parallel edges allowed. **Pseudograph:** loops allowed too.
- **Degree:** number of edges at a vertex (a loop counts twice).
- **Complete graph K_n:** every pair of vertices is adjacent; it has $n(n-1)/2$ edges.
- **Bipartite graph:** vertices split into two sets with every edge going between them. **$K_{m,n}$:** complete bipartite graph with mn edges.
- **Regular graph:** all vertices have the same degree.
- **Isomorphic graphs:** there is a bijection between vertex sets that preserves adjacency.
- **Homeomorphic graphs:** graphs that can be obtained from the same graph by subdividing edges (inserting degree-2 vertices).
- **Planar graph:** a graph that can be drawn in the plane with no edges crossing.
- **Euler path / circuit:** a path / closed path using every edge exactly once.
- **Hamiltonian path / cycle:** a path / cycle visiting every vertex exactly once.
- **Chromatic number $\chi(G)$:** the least number of colours needed so adjacent vertices get different colours.
- **Pigeonhole principle:** if more than n objects are put into n boxes, some box gets at least two.

💡 Core concepts

Handshaking theorem: $\Sigma \deg(v) = 2|E|$. So the number of odd-degree vertices is always **even**.

Representations: adjacency matrix, incidence matrix, adjacency list.

Euler and Hamilton

	Euler	Hamiltonian
Visits	Every edge once	Every vertex once
Circuit exists when	Connected and **every vertex has even degree**	No simple test; sufficient: **Dirac** (every degree ≥ n/2) or **Ore** (deg u + deg v ≥ n for non-adjacent u, v)
Path exists when	Connected and **exactly 0 or 2 odd-degree vertices**	

Planar graphs

- **Euler's formula:** for a connected planar graph, **$V - E + F = 2$** (F = faces, including the outer one).
- For a simple connected planar graph with $V \geq 3$: **$E \leq 3V - 6$**. If it has no triangles: **$E \leq 2V - 4$**.
- **Kuratowski's theorem:** a graph is planar exactly when it has no subgraph homeomorphic to **K_5** or **$K_{3,3}$**.

Bipartite test: a graph is bipartite exactly when it has **no odd cycle**, i.e. it is 2-colourable.

Graph colouring facts: $\chi(K_n) = n$; χ(bipartite with an edge) = 2; χ(cycle C_n) = 2 if n even, 3 if n odd; χ(any planar graph) ≤ 4 (four colour theorem). Applications: exam timetabling, register allocation, map colouring.

Isomorphism checks: isomorphic graphs must have the same number of vertices, edges, degree sequence, cycles of each length, and connectivity. If any of these differs they are **not** isomorphic; if all match, find an explicit mapping.

Counting techniques

Rule	Formula
Sum rule	Tasks done in m or n disjoint ways: $m + n$
Product rule	Task in steps of m and n ways: $m \times n$
Permutations	$P(n, r) = n! / (n - r)!$
Combinations	$C(n, r) = n! / (r!(n - r)!)$
With repetition	Arrangements n^r; selections $C(n + r - 1, r)$
Identical objects	$n! / (n_1!\, n_2! \ldots n_k!)$
Inclusion–exclusion	$\lvert A \cup B\rvert = \lvert A\rvert + \lvert B\rvert - \lvert A \cap B\rvert$

Generalized pigeonhole: if N objects go into k boxes, some box has at least $\lceil N/k \rceil$ objects.

🧮 Solved examples

Example 1 (K_5 and $K_{3,3}$ are not planar).

- K_5: V = 5, E = 10. Planar would need $E \leq 3V - 6 = 9$. Since $10 > 9$, **K_5 is non-planar**.
- $K_{3,3}$: V = 6, E = 9, and it has no triangles (bipartite). Planar would need $E \leq 2V - 4 = 8$. Since $9 > 8$, **$K_{3,3}$ is non-planar**.

Example 2 (Euler's formula). A connected planar graph has 10 vertices, each of degree 3. Edges = $(10 \times 3)/2 = 15$ (handshaking). Faces $F = 2 - V + E = 2 - 10 + 15 =$ **7**.

Example 3 (Euler path). A graph has degrees 2, 3, 3, 4, 2. Exactly two vertices have odd degree, so it has an **Euler path** (starting at one odd vertex and ending at the other) but **no Euler circuit**.

Example 4 (pigeonhole).

- Among any 13 people, at least two share a birth month (13 people, 12 months).
- How many students guarantee that at least 6 get the same grade out of 5 grades? Need $\lceil N/5 \rceil \geq 6 \Rightarrow N \geq 26$. **26 students.**

Example 5 (inclusion–exclusion). How many integers from 1 to 100 are divisible by 2 or 3? $\lfloor 100/2 \rfloor + \lfloor 100/3 \rfloor - \lfloor 100/6 \rfloor = 50 + 33 - 16 =$ **67**.

Example 6 (counting). Arrangements of the letters of "BANANA" = $6! / (3! \cdot 2! \cdot 1!) = 720 / 12 =$ **60**. Committees of 3 from 8 people = $C(8, 3) =$ **56**.

Results to remember

```
Σ deg(v) = 2|E|;  number of odd-degree vertices is even
Kn has n(n-1)/2 edges;  Km,n has mn edges
Euler circuit: all degrees even;  Euler path: 0 or 2 odd-degree vertices
Dirac: deg(v) ≥ n/2 for all v ⇒ Hamiltonian
Planar: V - E + F = 2;  E ≤ 3V - 6;  no triangles: E ≤ 2V - 4
K5 and K3,3 are non-planar (Kuratowski)
Bipartite ⇔ no odd cycle ⇔ 2-colourable;  planar ⇒ x ≤ 4
Pigeonhole: N objects, k boxes ⇒ some box has ⌈N/k⌉
```

⭐ Important questions

1. State and prove the handshaking theorem; show the number of odd-degree vertices is even.
2. Define Euler and Hamiltonian paths/circuits with examples, and the conditions for their existence. *Draw one graph that is Eulerian but not Hamiltonian and one the other way round.*
3. Define planar graphs; prove Euler's formula and show K_5 and $K_{3,3}$ are non-planar.
4. Check whether two given graphs are isomorphic. *Compare invariants in a table, then give the mapping or the mismatch.*
5. Define graph colouring and chromatic number; find χ for given graphs and mention applications.
6. State the pigeonhole principle (and generalized form) and solve problems on it; solve counting problems with permutations, combinations and inclusion–exclusion.

Last-hour checklist

Read only this in the last hour before the paper.

- ☐ Reflexive, symmetric, antisymmetric, transitive definitions.
- ☐ Equivalence relation → partition; partial order → Hasse diagram.
- ☐ Lattice types; M_3 and N_5; D_n join = lcm, meet = gcd.
- ☐ One-one, onto, bijection; counting functions.
- ☐ Boolean laws, De Morgan, consensus; K-map grouping rules.
- ☐ Truth table of →; contrapositive.
- ☐ Inference rules MP, MT, HS, DS with names.
- ☐ Quantifier negation; ∀ with →, ∃ with ∧.
- ☐ Group axioms; subgroup test; Lagrange's theorem and proof.
- ☐ Cyclic groups; generators of $\mathbb{Z}_n$; cosets.
- ☐ Ring, integral domain, field.
- ☐ Handshaking; Euler vs Hamiltonian conditions.
- ☐ $V - E + F = 2$; $E \le 3V - 6$; K_5, $K_{3,3}$.
- ☐ Chromatic numbers; pigeonhole $\lceil N/k \rceil$.

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Discrete Structures	Thomas Koshy	Elsevier	WorldCat
2	Discrete Mathematics and Its Applications (6th ed.)	Kenneth H. Rosen	McGraw-Hill	WorldCat
3	Discrete Mathematical Structures (5th ed.)	B. Kolman, R. C. Busby, S. C. Ross	Prentice Hall	WorldCat
4	Mathematics: A Discrete Introduction	E. R. Scheinerman	Brooks/Cole	WorldCat
5	Discrete and Combinatorial Mathematics (5th ed.)	R. P. Grimaldi	Addison-Wesley	WorldCat
6	Discrete Mathematics (Schaum's Outline)	Seymour Lipschutz	McGraw-Hill	WorldCat
7	Discrete Mathematical Structures with Applications to Computer Science	J. P. Tremblay, R. Manohar	McGraw-Hill	WorldCat
8	Graph Theory with Applications to Engineering and Computer Science	Narsingh Deo	PHI	WorldCat
9	Combinatorics: Theory and Applications	V. Krishnamurthy	East-West Press	WorldCat

Official syllabus: AKTU B.Tech 2nd year CSE syllabus, 2023-24 scheme (PDF)